

Practical No: 8

Name:- Danish Altaf Solkar

Enrolment No:- 24111590805

Class:- CO3KB

Subject:- CGR

Aim :- WAP for 2D Rotation

Program :-

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
#include<math.h>
void main(){
int gd=DETECT,gm;
int x1,x2,y1,y2,xn,yn;
double r11,r12,r21,r22,th;
clrscr();
printf("\nEnter x y co ordinates of 2 lines and point :- ");
scanf("%d%d%d%d",&x1,&y1,&x2,&y2);
initgraph(&gd,&gm,"C:\\TURBOC3\\BGI");
printf("\nOriginal line before rotation :- ");
line(x1,y1,x2,y2);
printf("\nEnter the angle of rotation :- ");
scanf("%f",&th);
r11=cos((th*3.1428)/180);
r12=sin((th*3.1428)/180);
r21=(-sin(th*3.1428)/180);
r22=cos((th*3.1428)/180);
xn=((x2*r11)-(y2*r12));
yn=((x2*r21)-(y2*r22));
printf("\nLine after rotation :- ");
setcolor(10);
line(x1,y1,xn,yn);
getch();
closegraph();
}
```

OUTPUT :-

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Program: TC

Enter x y co ordinates of 2 lines and point :- 200 220 250 300_

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Program:

Original line before rotation :-
Enter the angle of rotation :-



DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Program:

Original line before rotation :-
Enter the angle of rotation :- 10

Line after rotation :-

